

Cooling towers

The steam produced in a nuclear power plant must be cooled down after it passes through the turbines, so that it can enter the reactor core as cold water once again. More cold water is used to cool the steam, and in doing so becomes warm. It cools down in huge cooling towers, like these at the Novovoronezh nuclear power station in Russia.

There are currently more than **400 nuclear reactors** in operation around the globe.

ATOMIC ENERGY

NUCLEAR POWER

Nuclear power generates about 10 percent of the world's electricity. It works by harnessing the huge amount of energy stored in atoms (see page 177). This can be released by splitting apart the nucleus (center) of the atom in a reaction known as fission. Nuclear power is not a form of renewable energy, as the fuel it uses can be used up. However, its power stations do not produce harmful greenhouse gases, unlike others that burn fossil fuels such as coal, oil, and gas.

NUCLEAR FISSION

To release the power of the atom, a neutron is fired at the nucleus of an unstable atom, such as uranium. This splits the nucleus, releasing energy and shooting out more neutrons. They then split other nuclei, releasing even more energy and triggering a chain reaction.

